Competitive Exam Practice 8: I/O Techniques

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type; TF = True/False.

Q1 *[GATE CS 1987 Q1-viii, verified — GATEOverflow V3 p.65]* [**MCQ, 1 mark**] On receiving an interrupt from an I/O device, the CPU:

- (A) Halts for a predetermined time.
- (B) Hands over control of the address bus and data bus to the interrupting device.
- (C) Branches off to the interrupt service routine immediately.
- (D) Branches off to the interrupt service routine after completion of the current instruction.

Q2 *[GATE CS 1995 Q1.3, verified — GATEOverflow V3 p.65]* [**MCQ, 1 mark**] In a vectored interrupt:

- (A) The branch address is assigned to a fixed location in memory.
- (B) The interrupting source supplies the branch information to the processor through an interrupt vector.
- (C) The branch address is obtained from a register in the processor.
- (D) None of the above.

Q3 *[GATE CS 1998 Q1.20, verified — GATEOverflow V3 p.65]* [**MCQ, 1 mark**] Which of the following is true?

- (A) Unless enabled, a CPU will not be able to process interrupts.
- (B) Loop instructions cannot be interrupted till they complete.
- (C) A processor checks for interrupts before executing a new instruction.
- (D) Only level-triggered interrupts are possible on microprocessors.

Q4 *[GATE CS 1996 Q1.24, verified — GATEOverflow V3 p.59]* [**MCQ, 1 mark**] For the daisy-chain scheme of connecting I/O devices, which of the following statements is true?

- (A) It gives non-uniform priority to various devices.
- (B) It gives uniform priority to all devices.
- (C) It is only useful for connecting slow devices to a processor device.

(D) It requires a separate interrupt pin on the processor for each device.

Q5 *[GATE CS 1987 Q2a, verified — GATEOverflow V3 p.59]* [**TF, 1 mark**] State whether the following statement is TRUE or FALSE:

In a microprocessor-based system, if a bus (DMA) request and an interrupt request arrive simultaneously, the microprocessor attends first to the bus request.

Q6 *[GATE CS 1987 Q2b, verified — GATEOverflow V3 p.59]* [**TF, 1 mark**] State whether the following statement is TRUE or FALSE:

Data transfer between a microprocessor and an I/O device is usually faster in a memory-mapped-I/O scheme than in an I/O-mapped-I/O scheme.

Q7 *[GATE CS 1990 Q4-ii, verified — GATEOverflow V3 p.59]* [**TF, 2 marks**] State whether the following statement is TRUE or FALSE, with reason:

The data transfer between memory and I/O devices using programmed I/O is faster than interrupt-driven I/O.

Q8 *[Original, GATE-pattern]* [**NAT, 2 marks**] A 2 GHz CPU reads input from a keyboard producing 10 keystrokes per second using a `READWAIT` polling loop of 2 instructions per failed iteration. The number of polling iterations the CPU executes *between* two consecutive keystrokes is ____________.

Q9 *[Original, GATE-pattern]* [**MCQ, 1 mark**] Despite being skew-immune and self-adapting to any device speed, most high-speed processor buses use synchronous rather than asynchronous protocols primarily because:

(A) Synchronous buses require no shared bus clock.

(B) Every asynchronous transfer pays two full round-trip delays, which a fast, similar-speed synchronous bus never pays.

(C) Synchronous buses are immune to slow devices needing wait states.

(D) Asynchronous protocols cannot support more than one bus master.

Q10 *[Original, GATE-pattern]* [**MCQ, 1 mark**] In a DMA transfer, *cycle stealing* refers to:

(A) The CPU taking memory bus cycles away from the DMA controller.

(B) The DMA controller taking memory bus cycles the CPU would otherwise have used, because both compete for the same bus.

(C) The CPU executing one ISR invocation per stolen word.

(D) The bus arbiter granting the bus to the CPU in preference to the DMA controller.

Q11 *[Original, GATE-pattern]* [**MCQ, 2 marks**] An I/O processor (IOP) differs from a DMA controller in that it:

(A) Moves the same volume of data with less extra hardware.

(B) Executes a whole channel program of transfers, formats, and simple decisions independently, interrupting the CPU only when the entire program completes.

(C) Never interrupts the CPU at all.

(D) Does not need to become bus master via arbitration.